Double Angle Formulae

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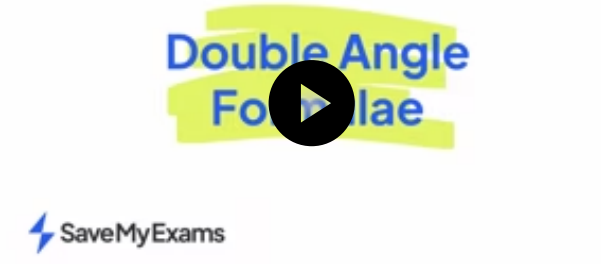

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Double angle formulae

What are the double angle formulae?

- The **double angle formulae** are derived from the 'A+B' versions of the **compound angle formulae**:
  - $\sin(2A) \equiv 2\sin A \cos A$
  - $\cos(2A) \equiv \cos^2 A - \sin^2 A$
    - $\cos(2A) \equiv 2\cos^2 A - 1$
    - $\cos(2A) \equiv 1 - 2\sin^2 A$
  - $\tan(2A) \equiv \dfrac{2\tan A}{1 - \tan^2 A}$

1. $\sin 2A \equiv 2\sin A \cos A$

$\sin(A+B) \equiv \sin A\cos B + \cos A\sin B$

LET B=A. THEN

$\sin(2A) = \sin(A+A)$
$= \sin A\cos A + \cos A\sin A$
$= 2\sin A\cos A$

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Help me understand

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

LET B=A. THEN

$$\cos(2A) \equiv \cos(A+A)$$
$$\equiv \cos A\cos A - \sin A\sin A$$
$$\equiv \cos^2 A - \sin^2 A$$

THE OTHER TWO FORMS COME FROM REARRANGING

$$\sin^2 A + \cos^2 A \equiv 1$$

AND SUBSTITUTING

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3. $\tan 2A \equiv \dfrac{2\tan A}{1-\tan^2 A}$

$$\tan(A+B) \equiv \frac{\tan A + \tan B}{1-\tan A\tan B}$$

LET B=A. THEN

$$\tan(2A) \equiv \tan(A+A)$$
$$\equiv \frac{\tan A + \tan A}{1-\tan A\tan A}$$
$$\equiv \frac{2\tan A}{1-\tan^2 A}$$

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Examiner Tips and Tricks

- The double angle formulae are **not** included in the formula booklet – you have to know them.
- If you forget them, you can always derive them from the compound angle formulae (which *are* in the formula booklet) as shown above.

Worked Example

written in the form

$$a\sin\theta(b\cos\theta + c) = 0$$

stating the values of a, b and c.

b) Hence solve, for $0 \le \theta < 360°$, the equation $7\sin 2\theta - 3\sin\theta = 0$. Give your answers to 1 decimal place.

a) $7\sin 2\theta = 7(2\sin\theta\cos\theta) = 14\sin\theta\cos\theta$

USE DOUBLE ANGLE FORMULA FOR sin

WE CAN REWRITE THE EQUATION AS

$14\sin\theta\cos\theta - 3\sin\theta = 0$

$\sin\theta(14\cos\theta - 3) = 0$

WITH $a=1$ $b=14$ $c=-3$

b) $\sin\theta(14\cos\theta - 3) = 0$

SO $\sin\theta = 0$ OR $\cos\theta = \frac{3}{14}$

IF $\sin\theta = 0$

$\theta = 0$ OR $\theta = 180$

YOU SHOULD KNOW THESE!

IF $\cos\theta = \frac{3}{14}$

$\theta = 77.63$ OR $\theta = 360 - 77.63 = 282.37$

USE CALCULATOR TO FIND PRINCIPAL VALUE

THEN FIND OTHER VALUES IN THE REQUIRED INTERVAL

THE SOLUTIONS ARE

$\theta = 0°, 77.6°, 180°, 282.4°$ (1 d.p.)

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